

Real Vector Space

A non-empty set V is said to form a real vector space (or a vector space over the field R) if

i) There is a binary composition $(+)$ on V , called addition, satisfying

(V1) $\alpha + \beta \in V$ for all $\alpha, \beta \in V$

(V2) $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in V$

(V3) $\alpha + (\beta + \gamma) = (\alpha + \beta) + \gamma$ for all $\alpha, \beta, \gamma \in V$

(V4) $\exists$ an element θ in V such that

$$\alpha + \theta = \alpha \text{ for all } \alpha \in V$$

(V5) For each α in V , $\exists$ an element $-\alpha$ in V such that $\alpha + (-\alpha) = \theta$

ii) There is an external composition of R with V , called multiplication by real numbers, satisfying

(V6) $c\alpha \in V$ for all $c \in R$, all $\alpha \in V$

(V7) $c(d\alpha) = (cd)\alpha$ for all $c, d \in R$, all $\alpha \in V$

(V8) $c(\alpha + \beta) = c\alpha + c\beta$ for all $c \in R$, all $\alpha, \beta \in V$

(V9) $(c+d)\alpha = c\alpha + d\alpha$, for all $c, d \in R$, all $\alpha \in V$

(V10) $1\alpha = \alpha$, 1 being the identity element in R .

Ex 1) Real Vector Space R^n .

Let V be the set of all ordered n tuples $\{(a_1, a_2, a_3, \dots, a_n) : a_i \in R\}$

Let $+$ be a composition defined on V by $(a_1, a_2, \dots, a_n) +$

$$(b_1, b_2, \dots, b_n) = (a_1 + b_1, a_2 + b_2, a_3 + b_3, \dots, a_n + b_n)$$

A scalar multiplication, is defined by $c(a_1, a_2, a_3, \dots, a_n)$

$$= (ca_1, ca_2, ca_3, \dots, ca_n), \quad c \in R.$$

$$V = R^n = \{(a_1, a_2, \dots, a_n) : a_i \in R\}$$

$$R^2 = \{(a_1, a_2) : a_i \in R\}$$

$$R^3 = \{(a_1, a_2, a_3) : a_i \in R\}$$

Ex-2) Real Vector Space $R_{m \times n}$.

Let V be the set of all real $m \times n$ matrices. Let $+$ be a composition on V called 'addition' defined by

$$(a_{ij})_{m \times n} + (b_{ij})_{m \times n} = (a_{ij} + b_{ij})_{m \times n}$$

and an external composition of R with V , called multiplication of matrices by real numbers defined by

$$c(a_{ij})_{m \times n} = (ca_{ij})_{m \times n} \quad c \in R$$

Check whether conditions $V_1 - V_{10}$ are satisfied.

Sol: All are satisfied

Ex-3) Real Vector Space P_n

Let V be the set of all real polynomials of degree less than n . A real polynomial 'f' of degree n is

$$f = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

where $a_0, a_1, a_2, \dots, a_n$ are real numbers.

Let $+$ be a composition on V , called addition, defined by

$$(a_0 + a_1x + a_2x^2 + \dots + a_mx^m) + (b_0 + b_1x + b_2x^2 + \dots + b_nx^n)$$

$$= (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + \dots + (a_m + b_m)x^m + b_{m+1}x^{m+1} + \dots + b_nx^n$$

$$= (a_0 + b_0) + (a_1 + b_1)x + \dots + (a_n + b_n)x^n \quad (n=m)$$

$$= (a_0 + b_0) + (a_1 + b_1)x + \dots + (a_n + b_n)x^n \quad (n=m)$$

and an external composition of R with V called multiplication of polynomial by real numbers is defined by

$$c(a_0 + a_1x + a_2x^2 + \dots + a_nx^n) = ca_0 + (ca_1)x + (ca_2)x^2 + \dots + (ca_n)x^n$$

Then V is a vector space ~~over~~ over R . $c \in R$,

Subspace:

Let V be a vector space over a field F under the operators addition '+' and external composition 'multiplication by scalars':

Theorem: A non-empty sub-set W of V is a subspace of V if

① $\alpha + \beta \in W$ for all $\alpha, \beta \in W$

② $c\alpha \in W$ $\forall \alpha \in W$ and all $c \in F$

$$V = R^2$$

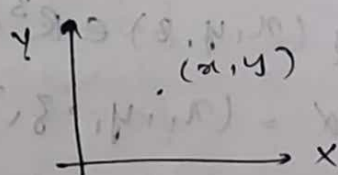
$$R^2 = \{(x, y) : x \in R, y \in R\}$$

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

$$\alpha = (x_1, y_1) \quad \beta = (x_2, y_2)$$

$$\alpha + \beta = (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$$

$$c\alpha = c(x_1, y_1) = (cx_1, cy_1)$$



Theorem: In a vector space V over a field F ,

① $0\alpha = \theta$ for all $\alpha \in V$

② $c\theta = \theta$ for all $c \in F$

③ $-1\alpha = -\alpha$ for all $\alpha \in V$,

1 being the identity element in F .

④ $c\alpha = \theta \Rightarrow$ either $c = 0$ or $\alpha = \theta$.

Ex) S be a subset of $\mathbb{R}^3$ defined by $S = \{(x, y, z) \in \mathbb{R}^3 : y = z = 0\}$

Then S is non empty because $(0, 0, 0) \in S$.

Let $\alpha, \beta \in S$

$$\alpha = (x_1, 0, 0) \quad \beta = (x_2, 0, 0)$$

$$\alpha + \beta = (x_1 + x_2, 0, 0) \in S$$

$$c\alpha = c(x_1, 0, 0) = (cx_1, 0, 0) \in S$$

S forms a subspace of V .

$$\textcircled{3} T = \{(x, y, z) \in \mathbb{R}^3 : x = z = 0\}$$

T is a subspace of $\mathbb{R}^3$

$$\textcircled{4} U = \{(x, y, z) \in \mathbb{R}^3 : \cancel{x = z = 0} \quad x = y = 0\}$$

U is a subspace of $\mathbb{R}^3$.

$\textcircled{5}$ Let S be a nonempty subset of $\mathbb{R}^3$ defined by

$$S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = z^2\}$$

$$\text{Let } \alpha = (x_1, y_1, z_1) \in S ; \beta = (x_2, y_2, z_2) \in S$$

$$x_1^2 + y_1^2 = z_1^2 ; x_2^2 + y_2^2 = z_2^2$$

$$\alpha + \beta = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$$

$\alpha + \beta$ may not belong to S because $(x_1 + x_2)^2 + (y_1 + y_2)^2$
may not be equal to $(z_1 + z_2)^2$

~~For~~ For example, let $\alpha = (3, -4, 5)$

$$\beta = (-3, 4, 5)$$

Then, $\alpha \in S ; \beta \in S$

$$\text{but } \alpha + \beta = (0, 0, 10) \notin S$$

So, S is not a subspace of $\mathbb{R}^3$

Let S be the subset defined by

$$S = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 1\}$$

$$(0, 0, 0) \notin S ; (0, 0, 1) \in S$$

Theorem: The intersection of two subspaces of a vector space ~~over~~ V over a field F is a subspace of V .

Proof: W_1, W_2

$$W_1 \cap W_2$$

$$i) \alpha, \beta \in W_1 \cap W_2 \Rightarrow \alpha + \beta \in W_1 \cap W_2 \quad \left(\begin{array}{l} \alpha, \beta \in W_1, \alpha, \beta \in W_2 \\ \alpha + \beta \in W_1, \alpha + \beta \in W_2 \end{array} \right)$$

$$ii) \alpha \in W_1 \cap W_2, c \in F \Rightarrow c\alpha \in W_1 \cap W_2$$
$$\left. \begin{array}{l} \alpha \in W_1, \alpha \in W_2 \\ c\alpha \in W_1, c\alpha \in W_2 \end{array} \right\} \nearrow$$

Note: 1) The intersection of a family of subspaces of V is a subspace of V .

2) The union of two subspaces of V is not, in general a subspace of V .

Example: $S = \{(x, y, z) \in \mathbb{R}^3 : y=0, z=0\}$

$$T = \{(x, y, z) \in \mathbb{R}^3, x=0, z=0\}$$

Let $\alpha = (1, 0, 0) \in S$; $\beta = (0, 1, 0) \in T$

Then $\alpha + \beta = (1, 1, 0)$

$$\alpha \in S \cup T; \beta \in S \cup T \text{ but } \alpha + \beta \notin S, \alpha + \beta \notin T$$

$$\Rightarrow \alpha + \beta \notin S \cup T$$

Hence, $S \cup T$ is not a subspace of $\mathbb{R}^3$.

Linear sum of two subspaces:

V is a vector space over a field F . U, W are two subspaces of V .

$$U + W = \{u + w : u \in U, w \in W\}$$

$$u \in U \Rightarrow u = \underbrace{u + 0}_{\in U + W}, \text{ where } 0 \in W$$

$$U \subset U + W$$

$$W \subset U + W$$

$$w \in W; w = 0 + w, \quad \left(\begin{array}{l} 0 \in U \\ w \in W \end{array} \right) \in U + W$$

Theorem 1: Let U, W be two subspaces of a vector space V over a field F , then $U+W$ is a subspace of V .

Theorem-2:

The subspace $U+W$ is the smallest subspace of V containing U and W .

Some important subspaces of a vector space:

1. Let $V(F)$ and let $\alpha \in V$. Then the set $W = \{c\alpha : c \in F\}$ forms a subspace of V .
2. Let $V(F)$ and let $\alpha, \beta \in V$. $W = \{c\alpha + d\beta : c \in F, d \in F\}$ forms a subspace of V .
3. Let $V(F)$ and let $\alpha_1, \alpha_2, \dots, \alpha_n \in V$, then the set $W = \{c_1\alpha_1 + c_2\alpha_2 + \dots + c_n\alpha_n : c_i \in F\}$ forms a subspace of V .

Proof: $0 \in W$

$$0 = 0\alpha \in W$$

Theorem 1

If $\alpha', \beta' \in W$

$$\alpha' = c_1\alpha \text{ for } c_1 \in F$$

$$\beta' = c_2\alpha \text{ for } c_2 \in F$$

$$\alpha' + \beta' = c_1\alpha + c_2\alpha = (c_1 + c_2)\alpha \in W$$

$$c'\alpha' = c'(c_1\alpha) = (c'c_1)\alpha \in W$$

Definition: Let $V(F)$, Let $\alpha_1, \alpha_2, \dots, \alpha_n \in V$.

A vector β in V is said to be a linear combination of the vectors $\alpha_1, \alpha_2, \dots, \alpha_n$ if β can be expressed as,

$$\beta = c_1\alpha_1 + c_2\alpha_2 + \dots + c_n\alpha_n$$

for some scalars $c_1, c_2, \dots, c_n$ in F .

Ex) $V(F)$ and $\alpha, \beta, \gamma \in V$

$$\alpha + \beta + \gamma; \alpha + 2\beta + 3\gamma; 0\alpha + \beta + 0\gamma$$

Ex) In $\mathbb{R}^3$, $\alpha = (4, 3, 5)$, $\beta = (0, 1, 3)$; $\gamma = (2, 1, 1)$,
 $\delta = (4, 2, 2)$

Examine if (i) α is a linear combo of β and γ

(ii) β is a linear combo of γ and δ

Soln: (i) Let $\alpha = c\beta + d\gamma$, where $c, d \in \mathbb{R}$

$$\text{Then } (4, 3, 5) = c(0, 1, 3) + d(2, 1, 1)$$

$$(4, 3, 5) = (0 + 2d, c + d, 3c + d)$$

$$\therefore 2d = 4; c + d = 3; 3c + d = 5$$

$$d = 2; c = 1; 3 + 2 = 5$$

α is a linear combo of β & γ

$$(0, 1, 3) = c(2, 1, 1) + d(4, 2, 2)$$

$$2c + 4d = 0 \quad \times$$

$$c + 2d = 1$$

Not a linear combo

$$\alpha = \beta + 2\gamma$$

* If S be a non-empty finite set $\{d_1, d_2, \dots, d_n\}$,
 then $L(S)$ is the set of all linear combinations
 of the vectors $d_1, d_2, \dots, d_n$.

$$S = \{d_1, d_2, \dots, d_n\}$$

$$L(S) = \{c_1 d_1 + c_2 d_2 + \dots + c_n d_n$$

a subspace of $V: c_i \in F\}$

Ex) Determine the subspace of $\mathbb{R}^3$ spanned by the
 vectors $\alpha = (1, 2, 3)$; $\beta = (3, 1, 0)$

Examine if $\gamma = (2, 1, 3)$, $\delta = (-1, 3, 6)$ are in the
 subspace.

$$c(1, 2, 3) + d(3, 1, 0)$$

$$\gamma = (2, 1, 3) = c(1, 2, 3) + d(3, 1, 0)$$

$$3c = 3 \Rightarrow \times$$

$$c + 3d = 2$$

$$2c + d = 1$$

$$d = 3/5, c = 1/5$$

$$\delta = (-1, 3, 6) = c(1, 2, 3) + d(3, 1, 0)$$

$$c + 3d = -1$$

$$2c + d = 3$$

$$3c = 6 \Rightarrow \checkmark$$

$$d = -1, c = 2$$

δ is in the subspace

Linear Dependence and Linear Independence:

A finite set of vectors $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ of a vector space V over a field is said to be linearly dependent if there exist scalars $c_1, c_2, \dots, c_n$ (not all zero) such that

$$c_1 \alpha_1 + c_2 \alpha_2 + \dots + c_n \alpha_n = \theta \quad \text{--- (i)}$$

The set is said to be linearly independent in V if relation (i) holds only when

$$c_1 = c_2 = \dots = c_n = 0$$

Ex) Examine if the set of vectors $\{(2, 1, 1), (1, 2, 2), (1, 1, 1)\}$ is linearly dependent in $\mathbb{R}^3$.

Soln: Let $\alpha = (2, 1, 1)$, $\beta = (1, 2, 2)$, $\gamma = (1, 1, 1)$

Consider the relation, $c_1 \alpha + c_2 \beta + c_3 \gamma = \theta$, where

c_1, c_2, c_3 are scalars.

$$c_1(2, 1, 1) + c_2(1, 2, 2) + c_3(1, 1, 1) = (0, 0, 0)$$

$$2c_1 + c_2 + c_3 = 0$$

$$c_1 + 2c_2 + c_3 = 0$$

$$c_1 + 2c_2 + c_3 = 0$$

$$\frac{c_1}{-1} = \frac{c_2}{-1} = \frac{c_3}{3} = k \Rightarrow c_1 = -k, c_2 = -k, c_3 = 3k$$

$$c_1 = -1, c_2 = -1, c_3 = 3$$

$$-\alpha - \beta + 3\gamma = \theta$$

The given vectors are linearly dependent.

Ex) Prove that the set of vectors $\{(1, 2, 2), (2, 1, 2), (2, 2, 1)\}$ is linearly independent in $\mathbb{R}^3$

Soln: Let $\alpha = (1, 2, 2)$, $\beta = (2, 1, 2)$ and $\gamma = (2, 2, 1)$

Consider the relation

$$c_1 \alpha + c_2 \beta + c_3 \gamma = \theta, \text{ where } c_1, c_2, c_3 \in \mathbb{R}$$

$$c_1(1, 2, 2) + c_2(2, 1, 2) + c_3(2, 2, 1) = (0, 0, 0)$$

$$\begin{cases} c_1 + 2c_2 + 2c_3 = 0 \\ 2c_1 + c_2 + 2c_3 = 0 \\ 2c_1 + 2c_2 + c_3 = 0 \end{cases} \Rightarrow \begin{cases} c_2 = c_3 \\ c_1 = -4c_2 \end{cases} \Rightarrow c_2 = c_1 = c_3 = 0$$

The set of vectors is linearly independent.

Definition and Dimension of a vector space:

$V(F)$ is said to be finite dimensional if $\exists$ a finite set of vectors in V generating V , otherwise V is said to be infinite dimensional.

Definition: Let $V(F)$. A set S of vectors in V is said to be a basis of V if

1) S is linearly independent in V .

2) S generates V .

$$V = \mathbb{R}^2; S = \{(1, 0), (0, 1)\}$$

$\epsilon_1 = (1, 0), \epsilon_2 = (0, 1)$

$$c_1 \epsilon_1 + c_2 \epsilon_2 = (0, 0) \Rightarrow c_1 = c_2 = 0$$

$$c_1(1, 0) + c_2(0, 1) = (0, 0)$$

$$(0, 0) \Rightarrow (c_1, c_2) = (0, 0)$$

$$\text{Let } \xi \in \mathbb{R}^2 \mid \xi = a\epsilon_1 + b\epsilon_2$$

$$\xi = (a, b) = a(1, 0) + b(0, 1) = a\epsilon_1 + b\epsilon_2$$

$$\xi \in L(S) \Rightarrow \xi = a\epsilon_1 + b\epsilon_2$$

$$\mathbb{R}^2 \subseteq L(S)$$

$$S \subset \mathbb{R}^2$$

$$L(S) \subset \mathbb{R}^2$$

$$L(S) = \mathbb{R}^2$$

Ex) The set $E = \{e_1 = (1, 0, 0), e_2 = (0, 1, 0), e_3 = (0, 0, 1)\}$ is a basis of $\mathbb{R}^3$. E is the std. basis of $\mathbb{R}^3$.

Ex) The set $E = \{e_1 = (1, 0, 0, \dots, 0), e_2 = (0, 1, 0, \dots, 0), \dots, e_n = (0, 0, 0, \dots, 0, 1)\}$ is a basis of $\mathbb{R}^n$.

E is the standard basis of $\mathbb{R}^n$.

Ex) Prove that the set $S = \{(1, 0, 1), (0, 1, 1), (1, 1, 0)\}$ is a basis of $\mathbb{R}^3$.

Solⁿ: let $d_1 = (1, 0, 1)$, $d_2 = (0, 1, 1)$, $d_3 = (1, 1, 0)$

$$c_1 d_1 + c_2 d_2 + c_3 d_3 = 0$$

$$c_1(1, 0, 1) + c_2(0, 1, 1) + c_3(1, 1, 0) = (0, 0, 0)$$

$$c_1 + c_3 = c_2 + c_3 = c_1 + c_2 = 0$$

$$\Rightarrow c_1 = c_2 = c_3 = 0 \text{ (linearly independent)}$$

let $\xi = (a, b, c)$ be an arbitrary vector of $\mathbb{R}^3$.

let us examine if $\xi \in L(S)$.

If possible, let $\xi = \lambda_1 d_1 + \lambda_2 d_2 + \lambda_3 d_3$, for real $\lambda_1, \lambda_2, \lambda_3$

$$(a, b, c) = \lambda_1(1, 0, 1) + \lambda_2(0, 1, 1) + \lambda_3(1, 1, 0)$$

$$\lambda_1 + \lambda_3 = a \quad \lambda_2 + \lambda_3 = b \quad \lambda_1 + \lambda_2 = c$$

This is a non-homogeneous system of operators in $\lambda_1, \lambda_2, \lambda_3$.

The coefficient of the determinant

$$\begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = -2 \neq 0 \Rightarrow \text{By Cramer's rule, there exists a unique solution for } \lambda_1, \lambda_2, \lambda_3$$

$$\Rightarrow \xi \in L(S)$$

$$\begin{array}{l|l} \mathbb{R}^3 \subseteq L(S) & L(S) \subseteq \mathbb{R}^3 \\ S \subseteq \mathbb{R}^3 & L(S) = \mathbb{R}^3 \end{array}$$

1) Examine whether the following sets of vectors are linearly independent or not: $\{(1, 3, 2), (2, 8, -2), (-1, 9, 2)\}$

Soln:

$$c_1(1, 3, 2) + c_2(2, 8, -2) + c_3(-1, 9, 2) = (0, 0, 0) \quad -3c_1$$

$$c_1 + 2c_2 - c_3 = 0 \Rightarrow c_3 = c_1 + 2c_2 \Rightarrow c_3 = \frac{5c_1}{3}$$

$$3c_1 + 8c_2 + 9c_3 = 0 \Rightarrow 3c_1 + 8c_2 + 9\left(\frac{5c_1}{3}\right) = 0 \Rightarrow c_1 = 0$$

$$2c_1 - 2c_2 + 2c_3 = 0 \Rightarrow c_2 = c_1 + c_3 \Rightarrow c_2 = \frac{4c_1}{3}$$

$$c_1 = c_2 = c_3 = 0$$

Given set of vectors are linearly independent.

2) Determine whether the vector $(6, 10, 2)$ is a linear combo of the vectors $(1, 3, 2), (2, 8, -1), (-1, 9, 2)$.

$$c_1 + 2c_2 - c_3 = 6$$

$$3c_1 + 8c_2 + 9c_3 = 10$$

$$2c_1 - c_2 + 2c_3 = 2$$

$$5c_2 - 4c_3 = 10 \quad c_1 = \frac{8}{7}$$

$$7c_2 + 8c_3 = 2 \quad c_3 = \frac{10}{7}$$

$$17c_2 = 22 \Rightarrow c_2 = \frac{22}{17}$$

$$24 + 176 + 90 = 290$$

$$25 + 2(12) - 1(-19) = 20$$

$$1 + 19 = 20$$

(Yes)

3) Let $S = \{(1, 2, 1), (1, 1, -1), (4, 5, -2)\}$. Determine whether the

vector is in $L[S]$: $(1, -3, 5)$

$$c_1 + 2c_2 + c_3 = 1$$

$$c_1 + c_2 - c_3 = -3$$

$$4c_1 + 5c_2 - 2c_3 = 5$$

$$3 - 4 + 1 = 0$$

$$c_2 + 2c_3 = 4$$

$$c_2 + 2c_3 = 17$$

No soln.

(No)

4) Which of the following subset $[S]$ form a basis for the given vector space V ?

i) $S = \{(1, 1, 1), (1, 2, 3), (-1, 0, 1)\}$, $V = \mathbb{R}^3$

ii) $S = \{1, x, (x-1)x, x(x-1)(x-2)\}$, $V = P_3$, be the space of all polynomial of degree at most 3.

$$\begin{array}{ccc} 1 & 1 & 1 \\ 1 & 2 & 3 \\ -1 & 0 & 1 \end{array}$$

$$2 - 4 + 2 = 0$$

$$C_1 + C_2 + C_3 = 0$$

$$C_2 = -2C_1$$

$$C_1 + 2C_2 + 3C_3 = 0$$

$$-4C_1 + 4C_1 = 0$$

$$C_1 = C_3$$

i) No

$$ii) \quad 1 \quad x \quad x^2 - x \quad x^3 - 3x^2 + 2x$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & -3 & 2 & 0 \end{bmatrix}$$

⊖

$$ax^3 - 2bx^2 + 2cx + d = 0$$

$$dx^3 + (c - 3d)x^2 + (b - c + 2d)x + a = 0$$

$$d = 0 \mid a = 0 \mid c = 0 \mid b = 0 \checkmark$$

Yes

Dimension:

The number of vectors in ~~the~~^a basis of a vector space V is said to be the dimension (or rank) of V and is denoted by $\dim V$.

The null space $\{0\}$ is said to be of dimension 0.

Ex) i) The dimension of $\mathbb{R}^2$ is 2, since $E = \{(1, 0), (0, 1)\}$ is a basis.

ii) The dimension of $\mathbb{R}^3$ is 3, since $E = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is a basis.

iii) The dimension of $\mathbb{R}^n$ is n , since $E = \{(1, 0, 0, \dots, 0), (0, 1, 0, \dots, 0), \dots, (0, 0, \dots, 1)\}$ is a basis.

iv) $R_{m \times n}$ - Dimension of a vector space formed by all possible real $m \times n$ matrices is " $m \times n$ " = " mn ", since

$E = \{E_{11}, E_{12}, E_{13}, \dots, E_{1n}, E_{21}, \dots, E_{mn}\}$, where E_{ij} is an $m \times n$ matrix having 1 as the ij th element and 0 elsewhere is a basis.

v) The dimension of the vector space P_n of all real polynomials in x of degree $< n$ is n . The set of polynomials

$\{1, x, x^2, \dots, x^{n-1}\}$ is a basis.

Ex) Consider the vector space of continuous real valued functions defined on the interval $\{x \mid -1 \leq x \leq 1\}$. Determine whether the functions $1, x, x^2$ are dependent or independent.

$$c_1 + c_2 x + c_3 x^2 = 0, \text{ where the equality is to hold everywhere in } [-1, 1]$$

$$x = 0, c_1 = 0$$

$$x = 1, c_1 + c_2 + c_3 = 0 \Rightarrow c_2 + c_3 = 0$$

$$x = -1, c_1 - c_2 + c_3 = 0 \Rightarrow c_2 = c_3 = 0$$

The functions are independent.

Alt: Since the functions $1, x, x^2$ are differentiable on the given interval, let $p(x) = c_1 + c_2 x + c_3 x^2 = 0$

$$p(0) = c_1 = 0$$

$$p'(0) = c_2 = 0$$

$$p''(0) = 2c_3 = 0$$

$$c_1 = c_2 = c_3 = 0$$

Independent.

Note: Consider the same interval $[-1, 1]$. f, g, h are defined as $f(x) = 1, x \in [-1, 1]$

$$g(x) = \begin{cases} 0, & -1 \leq x \leq 0 \\ x, & 0 \leq x \leq 1 \end{cases}$$

$$h(x) = \begin{cases} 0, & -1 \leq x \leq 0 \\ x^2, & 0 \leq x \leq 1 \end{cases}$$

These functions are independent.

Consider $c_1 f(x) + c_2 g(x) + c_3 h(x) = 0 = p(x)$

$$p(0) = c_1 = 0, p(1) = c_2 + c_3 = 0 \Rightarrow c_2 = c_3 = 0$$

$$p(1/2) = 1/2 c_2 + 1/4 c_3 = 0$$

f, g, h are independent.

$$[-1, 0] \quad \{x \mid -1 \leq x \leq 0\}$$

$$0f(x) + 1g(x) + 1h(x) = 0$$

$$\text{for } -1 \leq x \leq 0$$

Note 2) $f_1(x), f_2(x), \dots, f_k(x)$ real-valued functions are all defined and are differentiable $k-1$ times on the interval $\{x | a \leq x \leq b\}$.

The Wronskian of the set of functions is defined on the interval is the determinant

$$W(x) = \begin{vmatrix} f_1(x) & f_2(x) & \dots & f_k(x) \\ f_1'(x) & f_2'(x) & \dots & f_k'(x) \\ f_1''(x) & f_2''(x) & \dots & f_k''(x) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(k-1)}(x) & f_2^{(k-1)}(x) & \dots & f_k^{(k-1)}(x) \end{vmatrix}$$

Theorem: A set of real-valued functions $f_1(x), f_2(x), \dots, f_k(x)$, differentiable $k-1$ times in $x \in [a, b]$ are independent if $W(x_0) \neq 0$ at some point x_0 in the interval.

Rank of a matrix:

Let A be a non-zero matrix of order $m \times n$. The rank of A is defined to be the greatest positive integer r such that A has at least one non-zero minor of order r . The rank of a zero matrix A is defined to be 0.

* For a non-zero matrix of order $m \times n$ A ,

$$0 < \text{rank}(A) \leq \min(m, n)$$

* For a square matrix A of order n ,

$\text{rank } A < n$ if it is a singular matrix.

$\text{rank } A = n$ if it is a non-singular matrix.

Ex-1) Let $A = \begin{pmatrix} 1 & 0 & 3 \\ 4 & -1 & 5 \\ 2 & 0 & 6 \end{pmatrix}$ ii) $A = \begin{pmatrix} 2 & 3 & -1 & 1 \\ 3 & 0 & 4 & 2 \\ 6 & 9 & -3 & 3 \end{pmatrix}$

$$\text{iii)} \begin{pmatrix} 2 & 3 & 0 & 4 & -1 \\ 1 & 2 & 5 & -7 & 3 \\ 4 & 6 & 0 & 8 & -2 \\ 3 & 5 & 5 & -3 & 2 \end{pmatrix}$$

Sol: i) $|A| = 0$ $\begin{vmatrix} 1 & 0 \\ 4 & -1 \end{vmatrix} \neq 0 \therefore \text{Rank} = 2$

ii) $\begin{vmatrix} 2 & 3 \\ 3 & 0 \end{vmatrix} \neq 0 \therefore \text{Rank} = 2$

iii) Check and you'll get rank = 2

Elementary operations:

1. Interchange of two rows R_{ij}
2. Multiplication of a row by a non-zero element in $F(R)$
3. Addition of a scalar multiple of one row to another row $- R_j + CR_i$

Definition - 1

An $m \times n$ matrix A is said to be row reduced if

- a) The first non-zero element in a non-zero row is 1 (called the leading 1) and
- b) Each column containing the leading 1 of some row has all other elements, zero.

Ex) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

Definition - 2

An $m \times n$ matrix A is said to be a row-reduced echelon matrix (or a row echelon matrix) if

- a) A is row-reduced.
- b) There is an integer r ($0 \leq r \leq m$) such that, the first r rows of A are non-zero rows and the remaining rows (if there be any) are all zero rows.

c) For each non-zero row, if leading element of the row i occurs in the column k_i , then $k_1 < k_2 < \dots < k_n$

Ex)
$$\begin{pmatrix} 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Ex) Obtain a row reduced matrix which is row equivalent to

$$\begin{pmatrix} 0 & 0 & 2 & 2 & 0 \\ 1 & 3 & 2 & 4 & 1 \\ 2 & 6 & 2 & 6 & 2 \end{pmatrix}$$

Soln: Let us apply elementary row operations on the matrix

$$\begin{pmatrix} 0 & 0 & 2 & 2 & 0 \\ 1 & 3 & 2 & 4 & 1 \\ 2 & 6 & 2 & 6 & 2 \end{pmatrix} \xrightarrow{\frac{1}{2}R_1} \begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 1 & 3 & 2 & 4 & 1 \\ 2 & 6 & 2 & 6 & 2 \end{pmatrix} \xrightarrow{\begin{matrix} R_2 - 2R_1 \\ R_3 - 2R_1 \end{matrix}}$$

$$\begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 1 & 3 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xleftarrow{R_3 - 2R_2} \begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 1 & 3 & 0 & 2 & 1 \\ 2 & 6 & 0 & 4 & 2 \end{pmatrix}$$

Ex) Apply the elementary row operations to reduce the following matrix to a row echelon matrix.

$$\begin{pmatrix} 2 & 0 & 4 & 2 \\ 3 & 2 & 6 & 5 \\ 5 & 2 & 10 & 7 \\ 0 & 3 & 2 & 5 \end{pmatrix}$$

Ex) Obtain a row echelon matrix which is row equivalent to

$$\begin{pmatrix} 0 & 0 & 2 & 2 & 0 \\ 1 & 3 & 2 & 4 & 1 \\ 2 & 6 & 2 & 6 & 2 \\ 3 & 9 & 1 & 10 & 6 \end{pmatrix}$$

$$\begin{aligned} \text{Soln: } \begin{pmatrix} 2 & 0 & 4 & 2 \\ 3 & 2 & 6 & 5 \\ 5 & 2 & 10 & 7 \\ 0 & 3 & 2 & 5 \end{pmatrix} &\xrightarrow{\frac{1}{2}R_1} \begin{pmatrix} 1 & 0 & 2 & 1 \\ 3 & 2 & 6 & 5 \\ 5 & 2 & 10 & 7 \\ 0 & 3 & 2 & 5 \end{pmatrix} \xrightarrow{\substack{R_2-3R_1 \\ R_3-5R_1}} \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 2 & 0 & 2 \\ 0 & 2 & 0 & 2 \\ 0 & 3 & 2 & 5 \end{pmatrix} \xrightarrow{\substack{\frac{1}{2}R_2 \\ \frac{1}{2}R_3}} \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 3 & 2 & 5 \end{pmatrix} \\ &\xrightarrow{\substack{R_4-3R_2 \\ R_3-R_2}} \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 2 \end{pmatrix} \xrightarrow{\frac{1}{2}R_4} \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow{\substack{R_3+R_4 \\ R_4-R_2}} \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

$$\xrightarrow{R_1-2R_3} \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = R \quad (R \text{ is now echelon matrix})$$

$$\begin{aligned} \text{Soln: } \begin{pmatrix} 0 & 0 & 2 & 2 & 0 \\ 1 & 3 & 2 & 4 & 1 \\ 2 & 6 & 2 & 6 & 2 \\ 3 & 9 & 1 & 10 & 6 \end{pmatrix} &\xrightarrow{\substack{\frac{1}{2}R_1 \\ R_3-2R_2}} \begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 1 & 3 & 2 & 4 & 1 \\ 0 & 0 & -2 & -2 & 0 \\ 3 & 9 & 1 & 10 & 6 \end{pmatrix} \xrightarrow{\substack{R_3/2 \\ R_3+R_1 \\ R_4-3R_1}} \begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 1 & 3 & 2 & 4 & 1 \\ 0 & 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \\ &\xrightarrow{R_3-2R_2} \begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 1 & 3 & 2 & 4 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{\substack{\frac{1}{2}R_2 \\ R_2-R_1 \\ -\frac{2}{3}R_2}} \begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 2 & -3 \\ 1 & 3 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2-R_3} \begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 1 & 3 & 2 & 4 & 1 \\ 1 & 3 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \\ &\xrightarrow{R_3-2R_2} \begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 1 & 3 & 2 & 4 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

$$\xrightarrow{R_3-4R_2} \begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 3 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{\substack{R_{13} \\ R_{23}}} \begin{pmatrix} 1 & 3 & 0 & 2 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Theorem: If a row reduced echelon matrix R has r non-zero rows, then $\boxed{\text{Rank } R = r}$

* The rank of a matrix remains invariant under an elementary row operation.

Theorem: If a matrix A be row equivalent to a row echelon matrix having r non-zero rows, then

$$\boxed{\text{Rank } A = r}$$

Ex) Determine the rank of the matrix $A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 8 & 6 \\ 0 & 0 & 5 & 8 \\ 3 & 6 & 6 & 3 \end{pmatrix}$

Sol: Apply elementary row operations on A to reduce it to a row echelon matrix.

$$\begin{aligned} A &\xrightarrow{\substack{R_2 - 2R_1 \\ R_4 - 3R_1}} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 6 & 6 \\ 0 & 0 & 5 & 8 \\ 0 & 0 & 3 & 3 \end{pmatrix} \xrightarrow{\substack{1/6 R_2 \\ 1/3 R_4}} \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 5 & 8 \\ 0 & 0 & 3 & 3 \end{pmatrix} \xrightarrow{\substack{R_1 - R_2 \\ R_3 - 5R_2 \\ R_4 - 3R_2}} \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{1/3 R_3} \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\ &\xrightarrow{\substack{R_1 + R_3 \\ R_2 - R_3}} B = \begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

B is a row reduced echelon matrix and B has 3 non-zero rows. Therefore $\text{Rank } B = 3$. Since A is row equivalent to B , $\text{rank } A = 3$.

Row Space and Column Space of a matrix:

Let A be an $m \times n$ matrix over a field F . Each row of A is an n -tuple vector in F^n . Let the row vectors of A be denoted by $\alpha_1, \alpha_2, \dots, \alpha_m$.

Let the column vectors be denoted by $\bar{\alpha}_1, \bar{\alpha}_2, \dots, \bar{\alpha}_n$.

The row vectors of A generate a vector space which is said to be the row space of A . It is denoted by $R(A)$.

$R(A)$ is a subspace of F^n .

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \begin{matrix} \xleftarrow{\alpha_1} \\ \xleftarrow{\alpha_2} \\ \vdots \\ \xleftarrow{\alpha_m} \end{matrix}$$

$m \times n$

Ex) Let $A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix}$

The row space of A is the linear span of the vectors

$$\alpha_1 = (1, 2, 0), \alpha_2 = (0, 1, 3), \alpha_3 = (2, 0, 1)$$

Therefore the row space of A

$$= \{c_1 \alpha_1 + c_2 \alpha_2 + c_3 \alpha_3 : c_i \in \mathbb{R}\}$$

The set $\{\alpha_1, \alpha_2, \alpha_3\}$ is linearly independent. Therefore the row rank of A is 3.

The column space of A is the linear span of $\bar{\alpha}_1 = (1, 0, 2)$, $\bar{\alpha}_2 = (2, 1, 0)$, $\bar{\alpha}_3 = (0, 3, 1)$

$$C(A) = \{c_1 \bar{\alpha}_1 + c_2 \bar{\alpha}_2 + c_3 \bar{\alpha}_3 : c_i \in \mathbb{R}\}$$

The set $\{\bar{\alpha}_1, \bar{\alpha}_2, \bar{\alpha}_3\}$ is linearly independent. Therefore the column rank of A is 3.

Ex) Determine the row rank and the column rank of the matrix A and verify that the row rank of A = column rank of A , where

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 \\ 3 & 2 & 6 & 9 \\ 1 & 1 & 2 & 6 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 2 & -3 \\ 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Row echelon matrix.

System of m linear equations in n unknowns:

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$\vdots$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

$$x_1 = c_1, x_2 = c_2, \dots, x_n = c_n$$

$$(c_1, c_2, c_3, \dots, c_n) \in V_n(F)$$

$$AX = B$$

coefficient matrix of the system of equations

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{pmatrix}_{m \times n}$$

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}_{n \times 1}$$

$$B = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}_{m \times 1}$$

$$(A, b) = \bar{A}$$

A solution of the system can be considered as an n -tuple vector of $V_n(F)$.

$$\text{Ex) } x_1 + 2x_2 = 3$$

$$3x_1 + x_2 = 4$$

$$(x_1, x_2) = (1, 1) \text{ Unique sol}^n$$

$$x_1 + 2x_2 = 3$$

$$3x_1 + 6x_2 = 7$$

No soln exist

$$\text{Ex) } x_1 + 2x_2 - x_3 = 0$$

$$2x_1 + x_2 - 2x_3 = 0$$

$$(1, 0, 1) (2, 0, 2) (3, 0, 3)$$

$$K(1, 0, 1), K \in R$$

Infinite solns

$$\bar{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{pmatrix}$$

$\bar{A}$ is called augmented matrix of the system

Theorem: Let $AX = B$ and $RX = S$ be two equivalent systems and α be a solution of $AX = B$. Then α is also a solution of $RX = S$.

Ex) Solve the system of equations.

$$\begin{aligned} x_1 + x_2 &= 4 \\ x_2 - x_3 &= 1 \\ 2x_1 + x_2 + 4x_3 &= 7 \end{aligned}$$

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 2 & 1 & 4 \end{pmatrix}$$

$$X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$B = \begin{pmatrix} 4 \\ 1 \\ 7 \end{pmatrix}$$

$$\bar{A} = \begin{pmatrix} 1 & 1 & 0 & 4 \\ 0 & 1 & -1 & 1 \\ 2 & 1 & 4 & 7 \end{pmatrix}$$

Applying elementary row operations on $\bar{A}$.

$$\bar{A} \xrightarrow{R_3 - 2R_1} \begin{pmatrix} 1 & 1 & 0 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & -1 & 4 & -1 \end{pmatrix} \xrightarrow{\substack{R_1 - R_2 \\ R_3 + R_2}} \begin{pmatrix} 1 & 0 & 1 & 3 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 3 & 0 \end{pmatrix} \xrightarrow{\frac{1}{3}R_3} \begin{pmatrix} 1 & 0 & 1 & 3 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$\begin{cases} x_1 = 3 \\ x_2 = 1 \\ x_3 = 0 \end{cases}$$

$$\begin{pmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \xrightarrow{\substack{R_1 - R_3 \\ R_2 + R_3}} \begin{pmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Ex) Solve the system of equations.

$$x_1 + 3x_2 + x_3 = 0$$

$$2x_1 - x_2 + x_3 = 0$$

$$A = \begin{pmatrix} 1 & 3 & 1 & 0 \\ 2 & -1 & 1 & 0 \end{pmatrix} \xrightarrow{R_2 - 2R_1} \begin{pmatrix} 1 & 3 & 1 & 0 \\ 0 & -7 & -1 & 0 \end{pmatrix} \xrightarrow{R_2 - 2R_1} \begin{pmatrix} 1 & 3 & 1 & 0 \\ 0 & -7 & -1 & 0 \end{pmatrix} \xrightarrow{R_1 - 3R_2} \begin{pmatrix} 1 & 3 & 1 & 0 \\ 0 & 1 & \frac{1}{7} & 0 \end{pmatrix} \xrightarrow{-\frac{1}{7}R_2} \begin{pmatrix} 1 & 3 & 1 & 0 \\ 0 & 1 & \frac{1}{7} & 0 \end{pmatrix}$$

The given system is equivalent to

$$x_1 + \frac{4}{7}x_3 = 0$$

$$x_2 + \frac{1}{7}x_3 = 0$$

$$\Rightarrow K \left(-\frac{4}{7}, -\frac{1}{7}, 1 \right), K \in \mathbb{R}$$

Infinite sol's.

Ex) Solve, if possible the system of equations

$$x_1 + 2x_2 - x_3 = 10$$

$$-x_1 + x_2 + 2x_3 = 2$$

$$2x_1 + x_2 - 3x_3 = 2$$

$$\bar{A} = \begin{pmatrix} 1 & 2 & -1 & 10 \\ -1 & 1 & 2 & 2 \\ 2 & 1 & -3 & 2 \end{pmatrix} \xrightarrow{R_3 + R_2 - R_1} \begin{pmatrix} 1 & 2 & -1 & 10 \\ -1 & 1 & 2 & 2 \\ 0 & 0 & 0 & -6 \end{pmatrix} \xrightarrow{R_2 + R_1} \begin{pmatrix} 1 & 2 & -1 & 10 \\ 0 & 3 & 1 & 12 \\ 0 & 0 & 0 & -6 \end{pmatrix}$$

The given system is equivalent to

$$\begin{pmatrix} 1 & 0 & -5/3 & 2 \\ 0 & 1 & 1/3 & 4 \\ 0 & 0 & 0 & -6 \end{pmatrix} \xleftarrow{R_1 - 2R_2} \begin{pmatrix} 1 & 2 & -1 & 10 \\ 0 & 1 & 1/3 & 4 \\ 0 & 0 & 0 & -6 \end{pmatrix} \xleftarrow{R_2/3}$$

$$x_1 - 5/3 x_3 = 2$$

$$x_2 + 1/3 x_3 = 4$$

$$0 = -6 \rightarrow \text{Inconsistent}$$

No soln

Ex) Solve, if possible

$$x_1 + 2x_2 - x_3 = 10$$

$$-x_1 + x_2 + 2x_3 = 2$$

$$2x_1 + x_2 - 3x_3 = 8$$

$$x + 2y + z - 3w = 0$$

$$2x + 4y + 3z + w = 0$$

$$3x + 6y + 4z - 2w = 0$$

Infinite solns

$$c \left(\frac{5}{3}, -\frac{1}{3}, 1 \right) + (2, 4, 0)$$

$$\left(\frac{5}{3}c + 2, -\frac{1}{3}c + 4, c \right)$$

Infinite solns

$$\bar{A} = \begin{pmatrix} 1 & 2 & -1 & 10 \\ -1 & 1 & 2 & 2 \\ 2 & 1 & -3 & 8 \end{pmatrix} \xrightarrow{R_2 + R_1} \begin{pmatrix} 1 & 2 & -1 & 10 \\ 0 & 3 & 1 & 12 \\ 0 & 4 & -2 & 20 \end{pmatrix} \xrightarrow{R_2 - R_1} \begin{pmatrix} 1 & 2 & -1 & 10 \\ 0 & 1 & 1/3 & 4 \\ 0 & 4 & -2 & 20 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -3 & 8 \\ 0 & 0 & 10 & -12 \end{pmatrix} \xrightarrow{R_2 - 4R_3} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -3 & 8 \\ 0 & 4 & -2 & 20 \end{pmatrix} \xrightarrow{R_1 + 1/2 R_3} \begin{pmatrix} 1 & -2 & 1 & -10 \\ 0 & 1 & -3 & 8 \\ 0 & 4 & -2 & 20 \end{pmatrix} \xrightarrow{R_1 + 2R_2} \begin{pmatrix} 1 & 0 & -5 & 2 \\ 0 & 1 & -3 & 8 \\ 0 & 4 & -2 & 20 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -3 & 8 \\ 0 & 0 & 5 & -6 \end{pmatrix}$$

$$\bar{A} = \begin{pmatrix} 1 & 2 & -1 & 10 \\ -1 & 1 & 2 & 2 \\ 2 & 1 & -3 & 8 \end{pmatrix} \xrightarrow{R_3 + 2R_2} \begin{pmatrix} 1 & 2 & -1 & 10 \\ -1 & 1 & 2 & 2 \\ 0 & 3 & 1 & 12 \end{pmatrix} \xrightarrow{R_2 + R_1} \begin{pmatrix} 1 & 2 & -1 & 10 \\ 0 & 3 & 1 & 12 \\ 0 & 3 & 1 & 12 \end{pmatrix}$$

$$\xrightarrow{R_3 - R_2} \begin{pmatrix} 1 & 2 & -1 & 10 \\ 0 & 3 & 1 & 12 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 + R_2} \begin{pmatrix} 1 & 5 & 0 & 22 \\ 0 & 3 & 1 & 12 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$x_1 + 5x_2 = 22$$

$$3x_2 + x_3 = 12$$

$$x_3 = C$$

$$\left(\frac{5C}{3} + 2, \frac{12-C}{3}, C \right)$$

$$C \left(\frac{5}{3}, -\frac{1}{3}, 1 \right) + (2, 4, 0)$$

$$\bar{A} = \begin{pmatrix} 1 & 2 & 1 & -3 & 0 \\ 2 & 4 & 3 & 1 & 0 \\ 3 & 6 & 4 & -2 & 0 \end{pmatrix} \xrightarrow{R_3 - R_2 - R_1} \begin{pmatrix} 1 & 2 & 1 & -3 & 0 \\ 2 & 4 & 3 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 1 & -3 & 0 \\ 0 & 0 & 1 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_1 - R_2} \begin{pmatrix} 1 & 2 & 0 & -10 & 0 \\ 0 & 0 & 1 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$x + 2y - 10w = 0$$

$$(w = c) \quad (y = d)$$

$$8x + 7w = 0$$

$$\left(\frac{24c}{7}, -7c, k, c \right)$$

Infinite solⁿs

$$(10c - 2d, d, -7c, c)$$

Choosing $y = c$, $w = d$, where c and d are arbitrary real numbers, the solution

$$(-2c + 10d, c, -7d, d)$$

$$= c(-2, 1, 0, 0) + d(10, 0, -7, 1)$$

$$= c\alpha_1 + d\alpha_2$$

Note: The solutions form a vector space generated by the vectors $(-2, 1, 0, 0)$ and $(10, 0, -7, 1)$ which are linearly independent.

The rank of the solution space is therefore 2.

The rank of the matrix A is 2.

$$\text{Rank of } A + \text{Rank of } X(A) = 4 \text{ (The no. of unknowns)}$$

Theorem: Let $AX = 0$ be a homogenous system of n unknowns and $X(A)$ be the solution space of the system

$$\text{Then rank of } A + \text{rank of } X(A) = n.$$

Non-homogenous system:

Theorem: A necessary and sufficient condition for a non-homogenous system $AX = B$ to be consistent is

$$\text{Rank of } A = \text{Rank of } \bar{A}$$

where $\bar{A}$ is the augmented matrix of the system.

If the non-homogenous system $AX = B$ possesses a solution X_0 , then all solutions of the system are obtained by adding X_0 to the general solutions of the associated homogenous system.

$$\left. \begin{array}{l} AX_0 = B \\ AX = 0 \end{array} \right\} \begin{array}{l} AX_0 + AX = B \\ A(X_0 + X) = B \end{array}$$

Ex) solve, if possible

$$x + 2y + z - 3w = 1$$

$$2x + 4y + 3z + w = 3$$

$$3x + 6y + 4z - 2w = 5$$

$$\bar{A} = \begin{pmatrix} 1 & 2 & 1 & -3 & 1 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 6 & 4 & -2 & 5 \end{pmatrix} \xrightarrow{R_3 - R_1 - R_2} \begin{pmatrix} 1 & 2 & 1 & -3 & 1 \\ 2 & 4 & 3 & 1 & 3 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 1 & -3 & 1 \\ 0 & 0 & 1 & 7 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\xrightarrow{R_1 - R_2} \begin{pmatrix} 1 & 2 & 0 & -10 & 0 \\ 0 & 0 & 1 & 7 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Rank of $\bar{A} = 3$

Rank of $A = 2$

Rank of $(\bar{A} \neq A) \Rightarrow$ System is inconsistent.

Ex) $x + 2y + z - 3w = 1$

$$2x + 4y + 3z + w = 3$$

$$3x + 6y + 4z - 2w = 4$$

$$\bar{A} = \begin{pmatrix} 1 & 2 & 1 & -3 & 1 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 6 & 4 & -2 & 4 \end{pmatrix} \xrightarrow{R_3 - R_1 - R_2} \begin{pmatrix} 1 & 2 & 1 & -3 & 1 \\ 2 & 4 & 3 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 1 & -3 & 1 \\ 0 & 0 & 1 & 7 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Rank of $\bar{A} = 2$ / Rank of $A = 2$

System is consistent.

$$x + 2y - 10w = 0$$

$$z + 7w = 1$$

$$y = c / w = d$$

Solⁿ set: $(10d - 2c, c, 1 - 7d, d)$

$$= (0, 0, 1, 0) + c(-2, 1, 0, 0) + d(10, 0, -7, 1)$$

$AX = B$ is consistent iff $S = \{X \mid AX = B\}$

Rank of $A = \text{Rank of } \bar{A}$

$S = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ - Column vectors of matrix A

$(c_1, c_2, \dots, c_n)$ is a solution

$$c_1 \alpha_1 + c_2 \alpha_2 + c_3 \alpha_3 + \dots + c_n \alpha_n = B \rightarrow \textcircled{1}$$

$$T = \{\alpha_1, \alpha_2, \dots, \alpha_n, B\}$$

$$S \subset T \Rightarrow L(S) \subset L(T)$$

$$L(T) \subset L(S)$$

Using $\textcircled{1}$, we can say that each element of T is a linear combination of the vectors of S . Therefore

$$L(T) \subset L(S)$$

$$\Rightarrow L(S) = L(T)$$

$\Rightarrow$ Column space of $A = \text{The column space of } \bar{A}$

$\Rightarrow$ Column rank of $A = \text{column rank of } \bar{A}$

i.e., rank of $A = \text{rank of } \bar{A}$

Theorem: $AX = B$ has a solution X_0 , then all solutions of $AX = B$ are obtained by adding X_0 to the general solution of $AX = 0$

Proof: $S \equiv \text{Set of all solutions of } AX = B$

$T \equiv \text{Set of all solutions of } AX = 0$

$$\text{Let } Y \in T \Rightarrow AY = 0$$

$$A(Y + X_0) = AY + AX_0$$

$$= 0 + B = B$$

$$\Rightarrow Y + X_0 \in S$$

Existence and no. of solutions of the non-homogeneous system $AX = B$ (A - $m \times n$ matrix)

Case-1: $m = n$

System is consistent if

$$\text{Rank of } A = \text{Rank of } \bar{A}$$

Sub-case (i):

$$\text{Rank of } A = \text{Rank of } \bar{A} = n$$

A is non-singular and A^{-1} exists

$$\text{Solution is } X = A^{-1}B$$

Sub-case (ii)

$$\text{Rank of } A = \text{Rank of } \bar{A} < n$$

$AX = 0$ has infinitely many solutions.

$AX = B$ has infinitely many solutions.

Case-2: $m < n$

$$\text{Rank of } A = \text{Rank of } \bar{A} \leq m$$

$\times$ (Infinitely many solutions)

Case-3:

$$\text{Rank of } A = \text{Rank of } \bar{A} \leq n$$

Sub-case (i):

$$\text{Rank of } A = \text{Rank of } \bar{A} = n$$

Let $X(A)$ be the solution space of $AX = 0$

$$\text{Then } \boxed{\text{Rank of } X(A) + \text{Rank of } A = n}$$

$$\text{Rank of } X(A) = 0$$

$\Rightarrow AX = 0$ possesses only the zero solution.

$\Rightarrow AX = B$ has a unique solution.

Sub-case (ii) :

$$\text{Rank of } A = \text{Rank of } \bar{A} < n$$

$$AX = 0 : \text{ Infinite solutions}$$

$$AX = B : \text{ Infinite no. of solutions.}$$

Ex) $x + y + z = 1$

$$x + 2y - z = b$$

$$5x + 7y + az = b^2$$

i) Only one solution

ii) No solution

iii) Many solutions

Solⁿ: The system has a unique solution if the coefficient determinant is non-zero.

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & -1 \\ 5 & 7 & a \end{vmatrix}$$

$$= 2a + 7 - 5 - a - 3$$

$$= a - 1$$

i) If $a \neq 1$, system has only one solution.

If $a = 1$, then the system has either no solution or many solutions.

When $a = 1$, coefficient matrix A and augmented matrix $\bar{A}$ are

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -1 \\ 5 & 7 & 1 \end{pmatrix}$$

$$\bar{A} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & -1 & b \\ 5 & 7 & 1 & b^2 \end{pmatrix}$$

Applying elementary row operations on $\bar{A}$

$$\bar{A} \xrightarrow[R_3 - 5R_1]{R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & b-1 \\ 0 & 2 & -4 & b^2-5 \end{pmatrix} \xrightarrow[R_3 - 2R_2]{R_1 - R_2} \begin{pmatrix} 1 & 0 & 3 & 2-b \\ 0 & 1 & -2 & b-1 \\ 0 & 0 & 0 & b^2-2b-3 \end{pmatrix}$$

Rank of $A = 2$ / If $b^2 - 2b - 3 \neq 0$, Rank of $\bar{A} = 3$

(i) If $b^2 - 2b - 3 \neq 0$, system is inconsistent $\Rightarrow$ No solution.

(ii) If $b^2 - 2b - 3 = 0$, then rank of $\bar{A}$ = Rank of A = 2
 $\Rightarrow$ System is consistent $\Rightarrow$ Many solutions.

$$b^2 - 3b + b - 3 \Rightarrow b(b-3) + 1(b-3)$$

$$(b-3)(b+1)$$

(i) $a = 1, b \neq 3, -1 \rightarrow$ No solution.

(ii) $a = 1, b = -1$ (or) $a = 1, b = 3 \rightarrow$ Many solutions.

x) $x + 2y + z = 1$

$$3x + y + 2z = 3$$

$$x + 7y + 2z = 1$$

Ans: $\left(1 - \frac{3}{5}c, -\frac{1}{5}c, c\right)$, where $c \in \mathbb{R}$

$$(1, 0, 0) + c \left(-\frac{3}{5}, -\frac{1}{5}, 1\right)$$

Take $c = 5k$, it is arbitrary.

$$(1 - 3k, -k, 5k) = (1, 0, 0) + k(-3, -1, 5)$$

$$\bar{A} = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 3 & 1 & 2 & 3 \\ 1 & 7 & 2 & 1 \end{pmatrix} \xrightarrow[R_3 - R_1]{R_2 - 3R_1} \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & -5 & -1 & 0 \\ 0 & 5 & 1 & 0 \end{pmatrix} \xrightarrow[R_2]{R_3 + R_2} \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & -5 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow[R_1 - R_2]{R_2/5} \begin{pmatrix} 1 & 1 & 4/5 & 1 \\ 0 & 1 & 1/5 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 - R_2} \begin{pmatrix} 1 & 0 & 3/5 & 1 \\ 0 & 1 & 1/5 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Rank of $\bar{A} = 2$ / Rank of $A = 2$

$$X = A^{-1}(B)$$

A is singular:

$$= \frac{1}{-12 - 8 + 20}$$

Infinite solⁿs.

$$x + \frac{3}{5}z = 1$$

$$z = 5k$$

$$y + \frac{1}{5}z = 0$$

$$(1 - 3k, -k, 5k) = (1, 0, 0) + k(-3, -1, 5)$$

Linear mapping:

V, W be two vector spaces over the field F .
 $T: V \rightarrow W$ is said to be a linear transformation or a linear mapping if

(i) $T(\alpha + \beta) = T(\alpha) + T(\beta)$ for all $\alpha, \beta \in V$

(ii) $T(c\alpha) = c T(\alpha)$ for all $\alpha \in V, c \in F$

Note:

$T(c\alpha + d\beta) = c T(\alpha) + d T(\beta)$ for all $\alpha, \beta \in V, c, d \in F$

Ex-① $T: V \rightarrow V$ defined by $T(\alpha) = \alpha$ for all $\alpha \in V$.

(ii) Let P be the vector space of all real polynomials

The mapping $D: P \rightarrow P$

$Dp(x) = \frac{d}{dx} p(x), p(x) \in P$

(iii) Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$T(x_1, x_2, x_3) = (x_1, x_2, 0), (x_1, x_2, x_3) \in \mathbb{R}^3$

Let $\alpha = (x_1, x_2, x_3), \beta = (y_1, y_2, y_3) \in \mathbb{R}^3$

$\alpha + \beta = (x_1 + y_1, x_2 + y_2, x_3 + y_3)$

$T(\alpha + \beta) = (x_1 + y_1, x_2 + y_2, 0)$

$= (x_1, x_2, 0) + (y_1, y_2, 0)$

$= T(\alpha) + T(\beta)$

For a real no. c , $c\alpha = (cx_1, cx_2, cx_3)$

$T(c\alpha) = (cx_1, cx_2, 0)$

$= c(x_1, x_2, 0)$

$= c T(\alpha)$

② $T: V \rightarrow W$ $T(\alpha) = \theta'$, for $\alpha \in V$

θ' being the null vector in W

③ Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by $T(x_1, x_2, x_3) = (x_1+1, x_2+1, x_3+1)$
for $(x_1, x_2, x_3) \in \mathbb{R}^3$

let $\alpha \in (x_1, x_2, x_3), \beta \in$

(ii) let V be the vector space of all real valued functions continuous on $[a, b]$ &

let $T: V \rightarrow \mathbb{R}$ be defined by:

$$T(f) = \int_a^b f(t) dt, f \in V$$

Then T is a linear functional.

Theorem:

let V and W be vector spaces over a field F and $T: V \rightarrow W$ be a linear mapping. Then,

(i) $T(\theta) = \theta'$

(ii) $T(-\alpha) = -T(\alpha)$ for all $\alpha \in V$

$$T(\theta + \theta) = T(\theta)$$

$$\Rightarrow T(\theta) + T(\theta) = T(\theta)$$

$$\Rightarrow \{-T(\theta) + T(\theta)\} + T(\theta) = -T(\theta) + T(\theta)$$

$$\Rightarrow \theta' + T(\theta) = \theta' \Rightarrow T(\theta) = \theta'$$

Kernel of a linear mapping:

let V and W be vector spaces over a field F . let $T: V \rightarrow W$ be a linear mapping. The set of all vectors $\alpha \in V$ such that $T(\alpha) = \theta'$ is said to be the kernel of T .

$$\text{Ker } T = \{\alpha \in V : T(\alpha) = \theta'\}$$

Theorem:

⊛ $\text{Ker } T$ is a subspace of V

Case-1) $\text{Ker } T = \{0\}$

Then $\text{Ker } T$ is a subspace of V

Case-2) $\text{Ker } T \neq \{0\}$

Let $\alpha \in \text{Ker } T$, then $T(\alpha) = 0'$

Let $c \in F$, $T(c\alpha) = cT(\alpha)$

$$= c0' = 0'$$

$$\Rightarrow c\alpha \in \text{Ker } T$$

Let $\alpha, \beta \in \text{Ker } T$

$$T(\alpha) = 0'$$

$$T(\beta) = 0'$$

$$T(\alpha + \beta) = T(\alpha) + T(\beta)$$

$$= 0' + 0'$$

$$\text{or, } T(\alpha + \beta) = 0'$$

$$\Rightarrow \alpha + \beta \in \text{Ker } T$$

Theorem:

Let $T: V \rightarrow W$, T is injective iff $\text{Ker } T = \{0\}$

Let T is injective

Since $T(0) = 0'$,

0 is the only pre image of $0'$

$$\text{So } \text{Ker } T = \{0\}$$

($\Rightarrow$) Conversely, let $\text{Ker } T = \{0\}$

and α, β be two vectors of V such that

$$T(\alpha) = T(\beta) \text{ in } W$$

$$\text{Now, } 0' = T(\alpha) - T(\beta)$$

$$= T(\alpha - \beta) \text{ by } \alpha - \beta \in \text{Ker } T = \{0\}$$

$$\Rightarrow \alpha - \beta \in \text{Ker } T = \{0\} \Rightarrow \alpha = \beta$$

$$\Rightarrow \alpha = \beta$$

$$T(\alpha) = T(\beta) \Rightarrow \alpha = \beta \text{ i.e., } T \text{ is injective.}$$

$$T(\alpha + \beta) = T(\alpha) + T(\beta)$$

$$T(\alpha - \beta) = T(\alpha) + T(-\beta) = T(\alpha) - T(\beta)$$

$$= T(\alpha) - T(\beta)$$

theorem: Let V and W be vector spaces over a field F . Let $T: V \rightarrow W$ be a linear mapping such that $\text{Ker } T = \{0\}$. Then the images of a linearly independent set of vectors in V are linearly independent in W .

proof: Let $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n \in V$ be independent. Suppose $c_1 \alpha_1 + c_2 \alpha_2 + \dots + c_n \alpha_n = 0$. Then $T(c_1 \alpha_1 + c_2 \alpha_2 + \dots + c_n \alpha_n) = T(0) = 0$. But $T(c_1 \alpha_1 + c_2 \alpha_2 + \dots + c_n \alpha_n) = c_1 T(\alpha_1) + c_2 T(\alpha_2) + \dots + c_n T(\alpha_n) = 0$. Since $T(\alpha_1), T(\alpha_2), \dots, T(\alpha_n)$ are linearly independent in W , we have $c_1 = c_2 = \dots = c_n = 0$. Hence $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ are linearly independent in V .

$$\text{Let } \alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n \in V$$

independent.

$$T(\alpha_1), T(\alpha_2), \dots, T(\alpha_n) \in W$$

$$c_1 T(\alpha_1) + c_2 T(\alpha_2) + \dots + c_n T(\alpha_n) = 0 \Rightarrow T(c_1 \alpha_1 + c_2 \alpha_2 + \dots + c_n \alpha_n) = 0$$

$$\text{Ker } T = \{0\}$$

$$\Rightarrow c_1 \alpha_1 + c_2 \alpha_2 + \dots + c_n \alpha_n = 0$$

$$\text{implies } c_1 = c_2 = \dots = c_n = 0$$

2) A mapping $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is defined by

$$T(x_1, x_2, x_3) = (x_1 + x_2 + x_3, 2x_1 + x_2 + 2x_3, x_1 + 2x_2 + x_3),$$

$$(x_1, x_2, x_3) \in \mathbb{R}^3$$

Show that T is a linear mapping

Find $\text{Ker } T$ and dimension of $\text{Ker } T$.

Soln: $T(x_1 + y_1, x_2 + y_2, x_3 + y_3) = (x_1 + y_1 + x_2 + y_2 + x_3 + y_3, 2(x_1 + y_1) + x_2 + y_2 + 2(x_3 + y_3), (x_1 + y_1) + 2(x_2 + y_2) + (x_3 + y_3))$

$$= (x_1 + x_2 + x_3, 2x_1 + x_2 + 2x_3, x_1 + 2x_2 + x_3) + (y_1 + y_2 + y_3, 2y_1 + y_2 + 2y_3, y_1 + 2y_2 + y_3)$$

$$= T(x_1, x_2, x_3) + T(y_1, y_2, y_3)$$

~~$T(ax_1, bx_2, cx_3) = (ax_1 + bx_2 + cx_3, 2ax_1 + 2bx_2 + cx_3, ax_1 + 2bx_2 + cx_3)$~~

$$T(ax_1, ax_2, ax_3) = (ax_1 + ax_2 + ax_3, 2ax_1 + x_2 + 2ax_3, ax_1 + 2ax_2 + ax_3)$$

$$= a(x_1 + x_2 + x_3, 2x_1 + x_2 + 2x_3, x_1 + 2x_2 + x_3)$$

$$= a T(x_1, x_2, x_3)$$

$\therefore T$ is a linear mapping.

Let $\alpha = (x_1, x_2, x_3)$

$$T(\alpha) = 0 \Rightarrow x_1 + x_2 + x_3 = 0$$

$$2x_1 + x_2 + 2x_3 = 0$$

$$x_1 + 2x_2 + x_3 = 0$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 1 & 2 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$x_1 + x_3 = 0$$

$$x_2 = 0$$

$$\text{Ker } T = \{(c, 0, -c) : c \in \mathbb{R}\}$$

$$\text{Dimension of Ker } T = 1$$

$$\text{Ker } T = L\{(1, 0, -1)\}, K \in \mathbb{R}$$

$$= L\{\alpha\} \text{ where } \alpha = (1, 0, -1)$$

Ex-2) A linear mapping $T: \mathbb{R}^3 \rightarrow \mathbb{R}^4$ is defined by
 $T(x_1, x_2, x_3) = (x_2 + x_3, x_3 + x_1, x_1 + x_2, x_1 + x_2 + x_3)$
 for $(x_1, x_2, x_3) \in \mathbb{R}^3$

Find $\text{Ker } T$. Verify that $\{T(e_1), T(e_2), T(e_3)\}$ is linearly independent in $\mathbb{R}^4$, where

$$e_1 = (1, 0, 0), e_2 = (0, 1, 0), e_3 = (0, 0, 1)$$

Soln: Let $\alpha = (x_1, x_2, x_3) \in \text{Ker } T$.

$$\text{Then } T(\alpha) = (0, 0, 0, 0)$$

$$\Rightarrow (x_2 + x_3, x_3 + x_1, x_1 + x_2, x_1 + x_2 + x_3) = (0, 0, 0, 0)$$

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \xrightarrow{} \begin{pmatrix} -1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow \begin{cases} x_1 = x_2 \\ x_1 + x_3 = 0 \\ x_2 = x_3 \\ x_3 = 0 \end{cases}$$

$$(x_1, x_2, x_3) = (0, 0, 0)$$

$$\text{Ker } T = \{(0, 0, 0)\} = \{0\}$$

$$\text{Ker } T = \{\alpha \in \mathbb{R}^3 : T(\alpha) = 0\}$$

$$\{e_1, e_2, e_3\} \text{ in } \mathbb{R}^3$$

$$T(e_1) = T(1, 0, 0) = (0, 1, 1, 1)$$

$$T(e_2) = T(0, 1, 0) = (1, 0, 1, 1)$$

$$T(e_3) = T(0, 0, 1) = (1, 1, 0, 1)$$

$$c_1(0, 1, 1, 1) + c_2(1, 0, 1, 1) + c_3(1, 1, 0, 1) = 0$$

$$c_2 + c_3 = 0$$

$$c_1 + c_3 = 0$$

$$c_1 + c_2 = 0$$

$$c_1 = c_2 = c_3 = 0$$

$$c_1 + c_2 + c_3 = 0$$

$\Rightarrow \{T(e_1), T(e_2), T(e_3)\}$ is linearly independent.

Image of a linear mapping:

Let V, W be vector spaces over a field F . Let $T: V \rightarrow W$ be a linear mapping.

$$\text{Im } T = \{T(\alpha) : \alpha \in V\}.$$

Theorem: Let V and W be vector spaces over a field F .

Let $T: V \rightarrow W$ be a linear mapping. Then,

$\text{Im } T$ is a subspace of W .

Proof: Let $0, 0'$ be the null elements of V & W respectively.

Since, $T(0) = 0'$, $0' \in \text{Im } T$.

$\text{Im } T$ is a non-empty subset of W .

Case-1) $\text{Im } T = \{0'\}$

Case-2) $\text{Im } T \neq \{0'\}$

Let $\xi \in \text{Im } T$, then $\exists$ some α in V such that

$$\begin{aligned} T(\alpha) &= \xi \\ \Rightarrow T(c\alpha) &= cT(\alpha) = c\xi \\ \Rightarrow c\xi &\in \text{Im } T \text{ for all } c \in F. \end{aligned}$$

Let $\xi \in \text{Im } T$, $\eta \in \text{Im } T$. Then $\exists$ elements α, β such that $T(\alpha) = \xi$, $T(\beta) = \eta$.

$$T(\alpha + \beta) = T(\alpha) + T(\beta)$$

$$= \xi + \eta$$

$$\Rightarrow \xi + \eta \in \text{Im } T.$$

Theorem: Let V and W be vector spaces over a field F .

Let $T: V \rightarrow W$ be a linear mapping and

$\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ be a basis of V , then the vectors

$T(\alpha_1), T(\alpha_2), \dots, T(\alpha_n)$ generate $\text{Im } T$.

Ex) Find $\text{Im } T$ and its dimension, where T is linear mapping in (i) previous two examples (ii) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^4$ defined as

i) $\text{Im } T = \{T(\alpha) : \alpha \in \mathbb{R}^3\}$

Basis of $V: \{(1,0,0), (0,1,0), (0,0,1)\}$

Basis of $\text{Im } T: \{T(1,0,0), T(0,1,0), T(0,0,1)\}$

$= \{(1,2,1), (1,1,2), (1,2,1)\}$

$= \{(1,2,1), (1,1,2)\}$

Defn: If $\{d_1, d_2, d_3\}$ is a basis of the domain space $\mathbb{R}^3$, $\text{Im } T$ is the linear space of the vectors $T(d_1), T(d_2), T(d_3)$

$\{e_1 = (1,0,0), e_2 = (0,1,0), e_3 = (0,0,1)\} \rightarrow \text{Basis of } \mathbb{R}^3$

$\text{Im } T = L\{(1,2,1), (1,1,2)\}$ (linear span of $\dots$)

$\text{Dim of Im } T = 2 \quad / \quad \text{Dim of Ker } T = 1$

ii) $e_1 = (1,0,0), e_2 = (0,1,0), e_3 = (0,0,1)$

$T(e_1) = (0,1,1,1)$

$T(e_2) = (1,0,1,1)$

$T(e_3) = (1,1,0,1)$

$\text{Im } T = L\{(0,1,1,1), (1,0,1,1), (1,1,0,1)\}$

$\text{Dim of Im } T = 3 \quad / \quad \text{Dim of Ker } T = 0$

Theorem: Let $T: V \rightarrow W$ be a linear mapping

$\{d_1, d_2, \dots, d_n\}$ be a basis of V

$\Rightarrow \{T(d_1), T(d_2), \dots, T(d_n)\}$ generates $\text{Im } T$

Proof: Let $\xi \in \text{Im } T$

There exists an element α in V such that $T(\alpha) = \xi$

Now, $\alpha = c_1 d_1 + c_2 d_2 + \dots + c_n d_n$ for some scalars $c_i \in F$

$\xi = T(c_1 d_1 + c_2 d_2 + \dots + c_n d_n)$
 $= c_1 T(d_1) + c_2 T(d_2) + \dots + c_n T(d_n)$

$\text{Im } T$ is generated by $T(d_1), T(d_2), \dots, T(d_n)$

Corollary: If $\text{Ker } T = \{0\}$, ~~then~~ and $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is a basis of V , then $\{T(\alpha_1), T(\alpha_2), \dots, T(\alpha_n)\}$ will be a basis of $\text{Im } T$.

Nullity and Rank of a linear mapping:

$$T: V \rightarrow W$$

The dimension of $\text{Ker } T$ is called the nullity of T .

The dimension of $\text{Im } T$ is called the rank of T .

Theorem: Let V and W be vector spaces over a field F and T be a linear mapping $T: V \rightarrow W$ and V is finite dimensional. Then,

$$\dim \text{Ker}(T) + \dim \text{Im}(T) = \dim V$$

In other words,

$$\text{The nullity of } T + \text{Rank of } T = \dim V$$

Ex) Determine the linear mapping $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ which maps the basis vectors $(1, 0, 0), (0, 1, 0), (0, 0, 1)$ of $\mathbb{R}^3$ to the vectors $(1, 1), (2, 3), (3, 2)$ respectively

i) Find $T(1, 1, 0), T(6, 0, -1)$

ii) Find $\text{Ker } T$ and $\text{Im } T$

iii) Prove that T is not one to one, but onto.

Soln: Let $\xi = (x, y, z)$ be an arbitrary element of $\mathbb{R}^3$.

$$\xi = (x, y, z)$$

$$= x(1, 0, 0) + y(0, 1, 0) + z(0, 0, 1)$$

Since T is linear,

$$T(\xi) = xT(1, 0, 0) + yT(0, 1, 0) + zT(0, 0, 1)$$

$$= x(1, 1) + y(2, 3) + z(3, 2)$$

$$T(x, y, z) = ((x + 2y + 3z), (x + 3y + 2z))$$

$$T(1,1,0) = (1+2, 1+3) = (3,4)$$

$$T(6,0,-1) = (6-3, 6-2) = (3,4)$$

$$x + 2y + 3z = 0$$

$$x + 3y + 2z = 0$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 5 \\ 0 & 1 & -1 \end{pmatrix}$$

$$x + 5z = 0$$

$$y - z = 0$$

$$(-5c, c, c)$$

$$\text{Ker } T = L\{(-5c, c, c)\}$$

$$= L\{(-5, 1, 1)\}$$

$$\dim \text{Ker } T = 1$$

$$T(e_1) = (1,1) \quad T(e_2) = (2,3) \quad T(e_3) = (3,2)$$

$$\text{Im } T = L\{(1,1), (2,3), (3,2)\} = L\{(2,3), (3,2)\}$$

$$\dim \text{Im } T = 3 - 1 = 2$$

iii) From (i) it is clear that T is not one-to-one.

Since $\text{Ker } T \neq \{0\}$, T is not one-to-one.

Since $\text{Im } T$ is a subspace of $\mathbb{R}^2$ and $\dim \text{Im } T = 2$,

$$\text{Im } T = \mathbb{R}^2$$

$\Rightarrow T$ is onto.

Ex-2) Determine the linear mapping $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ which maps the basis vector $(0,1,1), (1,0,1), (1,1,0)$ of $\mathbb{R}^3$ to the vectors $(1,1,1), (1,1,1), (1,1,1)$ respectively.

Verify that $\dim \text{Ker } T + \dim \text{Im } T = 3$.

Ex.) Prove that the linear mapping $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$$T(x,y,z) = (x+y, y+z, z+x), (x,y,z) \in \mathbb{R}^3 \text{ is}$$

One-to-one and onto.

Soln: Let $(x,y,z) \in \text{Ker } T$

$$\Rightarrow T(x,y,z) = (0,0,0)$$

$$(x+y, y+z, z+x) = (0, 0, 0)$$

$$\Rightarrow x = y = z = 0$$

$$\text{Ker } T = \{0\} \Rightarrow \text{One-to-one}$$

$$\{T(1,0,0), T(0,1,0), T(0,0,1)\} = \{(1,0,1), (1,1,0), (0,1,1)\}$$

$$\dim \text{Im } T = 3$$

Tutorial class - 2

1) Using Gauss-Jordan elimination method, find the

inverse of $\begin{pmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{pmatrix}$

$$\left(\begin{array}{ccc|ccc} -1 & 1 & 2 & 1 & 0 & 0 \\ 3 & -1 & 1 & 0 & 1 & 0 \\ -1 & 3 & 4 & 0 & 0 & 1 \end{array} \right) \xrightarrow[R_1+R_2]{R_1+R_3} \left(\begin{array}{ccc|ccc} +2 & 0 & +3 & +1 & +1 & 0 \\ 3 & -1 & 1 & 0 & 1 & 0 \\ -1 & 3 & 4 & 0 & 0 & 1 \end{array} \right)$$

$$\xrightarrow{R_2+R_1+R_3} \left(\begin{array}{ccc|ccc} +2 & 0 & +3 & +1 & +1 & 0 \\ 0 & 2 & 2 & -1 & 0 & 1 \\ -1 & 3 & 4 & 0 & 0 & 1 \end{array} \right) \xrightarrow{R_3-R_1} \left(\begin{array}{ccc|ccc} 2 & 0 & 3 & 1 & 1 & 0 \\ 0 & 2 & 2 & -1 & 0 & 1 \\ -3 & 3 & 1 & -1 & -1 & 1 \end{array} \right)$$

$$\xrightarrow[R_2/2]{R_1/2} \left(\begin{array}{ccc|ccc} 1 & 0 & 3/2 & 1/2 & 1/2 & 0 \\ 0 & 1 & 1 & -1/2 & 0 & 1/2 \\ -3 & 3 & 1 & -1 & -1 & 1 \end{array} \right) \xrightarrow{R_3+3R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 3/2 & 1/2 & 1/2 & 0 \\ 0 & 1 & 1 & -1/2 & 0 & 1/2 \\ 0 & 3 & 1/2 & 1/2 & 5/2 & 1 \end{array} \right)$$

$$\xrightarrow[\frac{2}{5}R_3]{R_3-3R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 3/2 & 1/2 & 1/2 & 0 \\ 0 & 1 & 1 & -1/2 & 0 & 1/2 \\ 0 & 0 & -5/2 & 4/5 & 1/5 & -1/5 \end{array} \right) \xrightarrow[R_1-3/2R_3]{R_2-R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -7/10 & 3/10 & 3/10 \\ 0 & 1 & 0 & -13/10 & -1/5 & 7/10 \\ 0 & 0 & 1 & 4/5 & 1/5 & -1/5 \end{array} \right)$$

$$\text{Inverse matrix} = \begin{pmatrix} -0.7 & 0.2 & 0.3 \\ -1.3 & -0.2 & 0.7 \\ 0.8 & 0.2 & -0.2 \end{pmatrix} \checkmark$$

-7	+13	8	-1	2	3	-0.7	0.2	0.3
+2	-2	+2	-13	-2	7	-1.3	-0.2	0.7
3	+7	-2	8	2	-2	0.8	0.2	-0.2

Investigate for what values of λ and μ the simultaneous equations $x+y+z=6$, $x+2y+3z=10$, $x+2y+\lambda z=\mu$ have
 i) unique ii) no solution iii) infinite solutions.

If $T: R^3(R) \rightarrow R^3(R)$ is linear transformation such that $T(1,1,1) = (1,2)$, $T(1,2,1) = (3,4)$, $T(0,0,1) = (1,5)$. Find $T(x,y,z)$

Consider $T: R^4(R) \rightarrow R^5(R)$ is linear transformation such that $T(x,y,z,w) = (x, x+y, y, z, 0)$. Find rank and nullity of the transformation as well as basis of null space and a basis of range space.

Consider $T: P_n(R) \rightarrow P_{n-1}(R)$ defined by $T(f(x)) = f'(x)$.

Examine if T is linear, Find $\text{Ker } T$, $\text{Im } T$.

Solns.

$$\begin{pmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{pmatrix} \xrightarrow[R_3-R_1]{R_2-R_1} \begin{pmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-1 & \mu-6 \end{pmatrix} \xrightarrow[R_3-R_2]{R_1-R_2} \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{pmatrix}$$

$$\xrightarrow[R_1+R_3]{\lambda \neq 3} \begin{pmatrix} 1 & 0 & 0 & 2 + \frac{\mu-10}{\lambda-3} \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & \frac{\mu-10}{\lambda-3} \end{pmatrix} \xrightarrow{R_2-2R_3} \begin{pmatrix} 1 & 0 & 0 & 2 + \frac{\mu-10}{\lambda-3} \\ 0 & 1 & 0 & 4 - 2\left(\frac{\mu-10}{\lambda-3}\right) \\ 0 & 0 & 1 & \frac{\mu-10}{\lambda-3} \end{pmatrix}$$

Rank of $A = 3$

$\lambda \neq 3 \Rightarrow$ unique solution.

$\lambda = 3, \mu \neq 10 \Rightarrow$ No solution

$\lambda = 3, \mu = 10 \Rightarrow$ Infinite solutions.

~~Answer~~ $(a_1x_1 + b_1x_2 + c_1x_3, a_2x_1 + b_2x_2 + c_2x_3)$

$$\begin{array}{l|l|l} a_1 + b_1 + c_1 = 1 & a_1 + 2b_1 + c_1 = 3 & c_1 = 1 \\ a_2 + b_2 + c_2 = 2 & a_2 + 2b_2 + c_2 = 4 & c_2 = 5 \\ a_1 + b_1 = 0 & b_1 = 2 & a_1 = -2 & c_1 = 1 \\ a_2 + b_2 = -3 & b_2 = 2 & a_2 = -5 & c_2 = 5 \end{array} \quad T(x,y,z) = \begin{pmatrix} -2x + 2y + z \\ -5x + 2y + 5z \end{pmatrix}$$

$$T: V \rightarrow W$$

$$\dim V = n \quad \dim W = m$$

$$\{\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n\} \quad \{\beta_1, \beta_2, \dots, \beta_m\}$$

A basis for V

A basis for W

Each $T(\alpha_i)$ is a linear combination of $\beta_1, \beta_2, \dots, \beta_m$

$$\text{Let } T(\alpha_1) = a_{11}\beta_1 + a_{21}\beta_2 + \dots + a_{m1}\beta_m$$

$$T(\alpha_2) = a_{12}\beta_1 + a_{22}\beta_2 + \dots + a_{m2}\beta_m$$

$$T(\alpha_n) = a_{1n}\beta_1 + a_{2n}\beta_2 + \dots + a_{mn}\beta_m$$

Let $\xi = x_1\alpha_1 + x_2\alpha_2 + \dots + x_n\alpha_n$ be an arbitrary vector of V

$$\text{and let } T(\xi) = y_1\beta_1 + y_2\beta_2 + \dots + y_m\beta_m \rightarrow \textcircled{1}$$

$$T(\xi) = T(x_1\alpha_1 + x_2\alpha_2 + \dots + x_n\alpha_n)$$

$$= x_1 T(\alpha_1) + x_2 T(\alpha_2) + \dots + x_n T(\alpha_n)$$

$$= x_1 (a_{11}\beta_1 + a_{21}\beta_2 + \dots + a_{m1}\beta_m) + x_2 (a_{12}\beta_1 + a_{22}\beta_2 + \dots + a_{m2}\beta_m)$$

$$+ \dots + x_n (a_{1n}\beta_1 + a_{2n}\beta_2 + \dots + a_{mn}\beta_m)$$

$$T(\xi) = (x_1 a_{11} + x_2 a_{12} + \dots + x_n a_{1n}) \beta_1$$

$$+ (x_1 a_{21} + x_2 a_{22} + \dots + x_n a_{2n}) \beta_2$$

$$+ (x_1 a_{m1} + x_2 a_{m2} + \dots + x_n a_{mn}) \beta_m \rightarrow \textcircled{2}$$

From $\textcircled{1}$ & $\textcircled{2}$,

$$y_1\beta_1 + y_2\beta_2 + \dots + y_m\beta_m = (\quad) \beta_1 + (\quad) \beta_2 + \dots + (\quad) \beta_m$$

$$y_1 = a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n$$

$$y_2 = a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n$$

$$y_m = a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n$$

$$Y = AX$$

where $Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix}$, $X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$, $A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$

Ex) A linear mapping $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is defined by $T(x_1, x_2, x_3) = (3x_1 + 2x_2 + x_3, x_1 + 3x_2 + 2x_3)$. Find the matrix of T relative to the ordered basis.

- i) $\{(1,0,0), (0,1,0), (0,0,1)\}$ of $\mathbb{R}^3$ and $\{(1,0), (0,1)\}$ of $\mathbb{R}^2$
 ii) $\{(0,1,0), (1,0,0), (0,0,1)\}$ of $\mathbb{R}^3$ and $\{(0,1), (1,0)\}$ of $\mathbb{R}^2$
 iii) $\{(0,1,1), (1,0,1), (1,1,0)\}$ of $\mathbb{R}^3$ and $\{(1,0), (0,1)\}$ of $\mathbb{R}^2$

Soln: i) $T(1,0,0) = (3,1) = 3(1,0) + 1(0,1)$

$T(0,1,0) = (-2,-3) = -2(1,0) - 3(0,1)$

$T(0,0,1) = (1,-2) = 1(1,0) - 2(0,1)$

The matrix of T is $\begin{pmatrix} 3 & -2 & 1 \\ 1 & -3 & -2 \end{pmatrix}$

ii) $T(0,1,0) = (-2,-3) = -3(0,1) - 2(1,0)$

$T(1,0,0) = (3,1) = 1(0,1) + 3(1,0)$

$T(0,0,1) = (1,-2) = -2(0,1) + 1(1,0)$

The matrix of T is $\begin{pmatrix} -3 & 1 & -2 \\ -2 & 3 & 1 \end{pmatrix}$

iii) $T(0,1,1) = (3,5) = 3(1,0) + 5(0,1) = (0,1,1)T$

$T(1,0,1) = (4,3) = 4(1,0) + 3(0,1) = (1,0,1)T$

$T(1,1,0) = (5,4) = 5(1,0) + 4(0,1) = (1,1,0)T$

The matrix of T is $\begin{pmatrix} 3 & 4 & 5 \\ 5 & 3 & 4 \end{pmatrix}$

Ex) The matrix of a linear mapping $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ relative to the ordered basis $\{(0,1,1), (1,0,1), (1,1,0)\}$ of $\mathbb{R}^3$ and $((1,0), (1,1))$ of $\mathbb{R}^2$ is $\begin{pmatrix} 1 & 2 & 4 \\ 2 & 1 & 0 \end{pmatrix}$

Find the matrix of T relative to the ordered basis $((1,1,0), (1,0,1), (0,1,1))$ of $\mathbb{R}^3$ and $((1,1), (0,1))$ of $\mathbb{R}^2$

Soln: $T(0,1,1) = 1(1,0) + 2(0,1) = (3,2)$

$T(1,0,1) = 2(1,0) + 1(0,1) = (3,1)$

$T(1,1,0) = 4(1,0) + 0(0,1) = (4,0)$

let $(x,y,z) \in \mathbb{R}^3$ and let $(x,y,z) = c_1(0,1,1) + c_2(1,0,1) + c_3(1,1,0)$

$(x,y,z) = (c_2 + c_3, c_1 + c_3, c_1 + c_2)$

$\Rightarrow c_2 + c_3 = x \Rightarrow c_1 = \frac{1}{2}(y + z - x)$

$c_1 + c_3 = y$

$c_2 = \frac{1}{2}(z + x - y)$

$c_1 + c_2 = z$

$c_3 = \frac{1}{2}(x + y - z)$

$T(x,y,z) = c_1 T(0,1,1) + c_2 T(1,0,1) + c_3 T(1,1,0)$
 $= c_1(3,2) + c_2(3,1) + c_3(4,0)$
 $= (3c_1 + 3c_2 + 4c_3, 2c_1 + c_2)$

$T(x,y,z) = (2x + 2y + z, \frac{1}{2}(-x + y + 3z)) \quad (x,y,z) \in \mathbb{R}^3$

$T(1,1,0) = (4,0) = 4(1,1) - 4(0,1) = (1,1,0)T$ (iii)

$T(1,0,1) = (3,1) = 3(1,1) - 2(0,1) = (1,0,1)T$

$T(0,1,1) = (3,2) = 3(1,1) - 1(0,1) = (0,1,1)T$

The matrix of T is $\begin{pmatrix} 4 & 3 & 3 \\ -4 & -2 & -1 \end{pmatrix}$